Session Objectives

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College of Osteopathic Medicine

1. Describe **cerebrovascular pathology**, **including ischemic stroke**, **hemorrhagic stroke**, **transient ischemic attack**, and **brain trauma.**
2. Outline the **histological changes in ischemic infarction** based on the time since the ischemic event.
3. Explain the **etiologies and classifications of ischemic strokes**, including **thrombotic**, **embolic**, and **global hypoperfusion**.
4. Classify and differentiate **hemorrhagic strokes**:
    1. Intracerebral hemorrhage (ICH): **Hypertension vs. Cerebral Amyloid Angiopathy (CAA).**
    2. Subarachnoid hemorrhage (SAH): **Traumatic, Aneurysmal, and Arteriovenous Malformations (AVMs)**.
5. Identify and describe the pathology of cerebrovascular trauma, including **brain herniation**, **skull fractures**, **concussion**, **subdural**, and **epidural hematomas**.
6. Discuss and describe the pathogenesis and morphological principles of direct parenchymal injuries, including **contusion**, **coup-contrecoup injury**, **diffuse axonal injury.**

Cerebrovascular disease (**CVD**)

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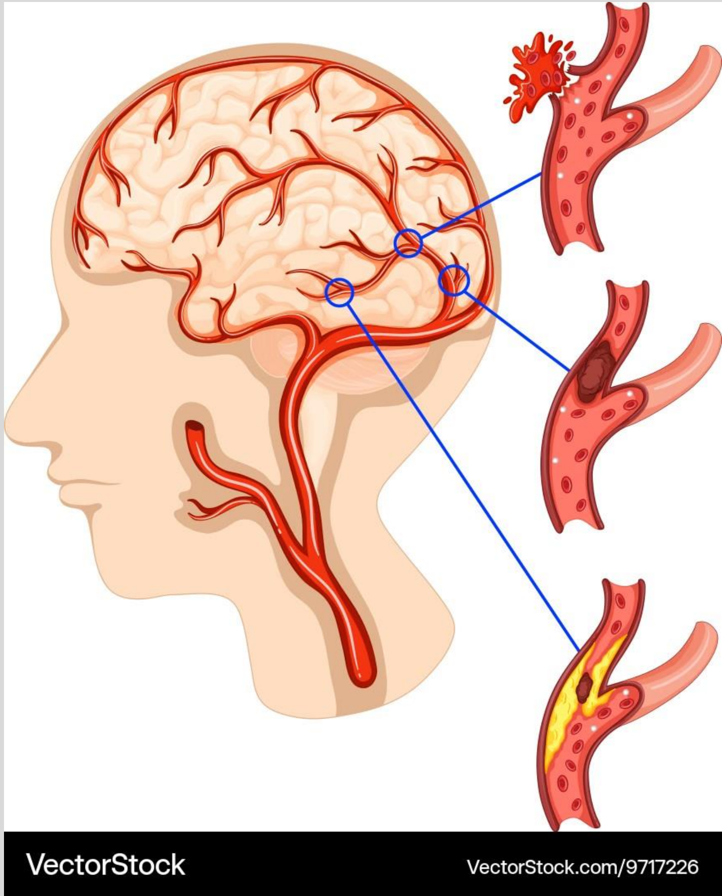
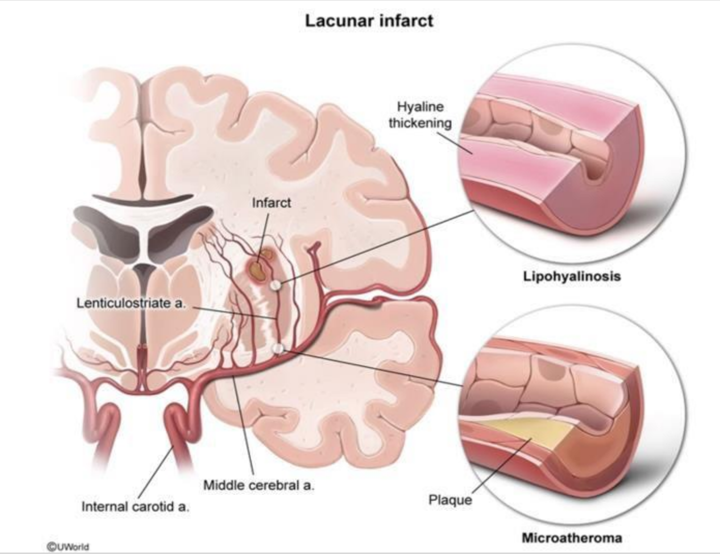
- Common is US
- **All disorders** with an area of the **brain temporarily or permanently affected by ischemia or bleeding** and one or more **cerebral blood vessels are involved** in the pathological processes
- **AKA Stroke** = Brain blood **flow** suddenly reduced with cell hypoxia due to **hypoxemia** (low blood O2) or **ischemia** (low blood flow) causes cell disability and **death/infarction**:
- When blood to the brain is reduced, tissue **survival depends** on **collateral** circulation presence, ischemia **duration** (sustained → infarction) & **magnitude** and **rapidity** of reduction
- Collateral flow source is **circle of Willis**, external carotid-ophthalmic artery collaterals - **Little if any** collateral flow for thalamus, basal ganglia & deep white matter deep penetrating vessels

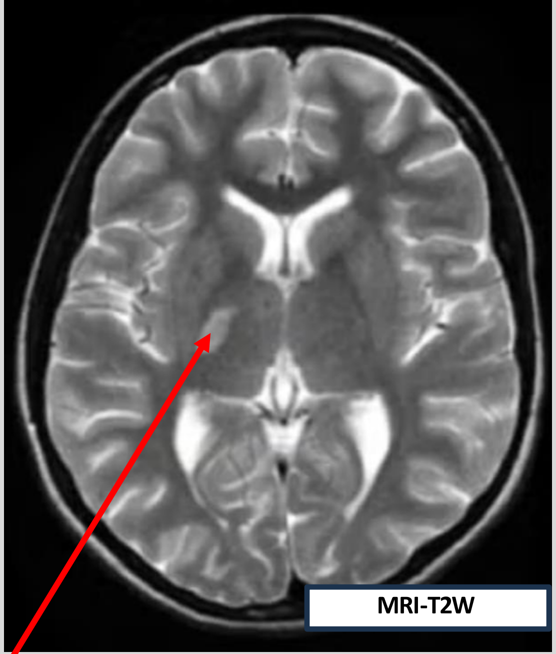


Thrombotic ischemic stroke: Lipohyalinosis

- Pathological process affects **small arteries** (arteriosclerosis) **and arterioles** (arteriolosclerosis), associated with **hypertension & diabetes**, can lead to **lacunar strokes**
- Arteries thicken/ narrow due to **fatty (lipo**) and **glassy/proteinaceous (hyaline)** material: Ex. Lenticulostriate arteries accumulation of lipid and hyaline material have thickened arterial walls → narrowing and potential occlusion
- **Chronic hypertensive vasculopathy** → lipohyalinosis of the small vessels → occlusion of small, penetrating arteries → lacunar stroke (**basal ganglia & subcortical white matter most prominent**)



Robbins and Cotran Pathologic Basis of Disease, 10th ed. fig. 28.16 *Lacunar infarcts* in the caudate and putamen (*arrows*)



- Named for the characteristic "**lake-like**" appearance of the resulting cavities in the brain tissue on MRI of the brain
- Axial *T2-weighted brain MRI* showing hyperintense signals in the basal ganglia/internal capsule region and other deep brain structures (**hyperintensity causes include stroke**, demyelination, inflammation, or tumors)
- Commonly **affected regions**: Internal capsule, basal ganglia, pons, corona radiata
- Depending on location, clinically **silent or severe impairment**

Hemorrhagic stroke pathophysiology

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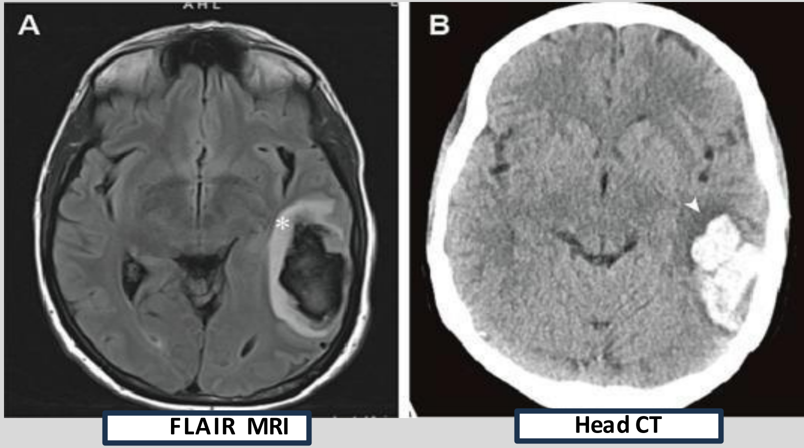
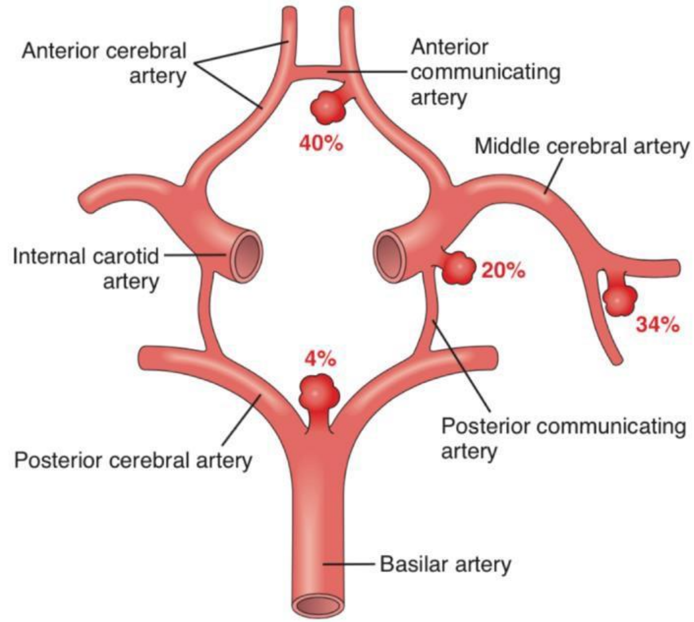
- Spontaneous (nontraumatic) intraparenchymal hemorrhages most common incidence ~60 years of age
- Hemorrhagic infarctions parallel ischemic with blood extravasation and resorption
- **Vessel rupture**
  - Hypertension, CAA, trauma, aneurysm or vascular malformations
  - **Blood extravasates** into brain parenchyma or subarachnoid space
- **Mass effect & edema** (see, asterix & arrowhead)
  - Hematoma **compresses surrounding brain** (hyperdense area on **CT**, usually in brain tissue)
  - Results in **elevated intracranial pressure** (ICP)
- Acute hemorrhages have central core of clotted blood that compresses adjacent parenchyma → secondary infarction of the affected brain tissue, with anoxic neuronal and glial changes as well as edema (eventually resolves) → hemosiderin and lipid-laden macrophages appear & proliferation of reactive astrocytes at periphery
- **Secondary ischemia & stroke**
  - **Adjacent vessels constrict** → worsens tissue hypoxia with **secondary ischemic strokes**
- **Toxic effects of blood products**
  - Hemoglobin breakdown releases free iron and reactive oxygen species (**ROS**) → cellular injury, BBB disruption, and neuroinflammation



- Neurobiol Dis. 2023 Jan:176:105948
- Brain **edema** formation and therapy after ICH
- (A) ICH patient's 2 days MRI FLAIR after hemorrhage showing significant **hyperintense perihematomal edema** (white asterisk) & lateral ventricle compression (B) ICH patient's 1 hour CT after hemorrhage showing **low-density brain edema** region (white arrow) around the hematoma

SAH: Spontaneous - aneurysm

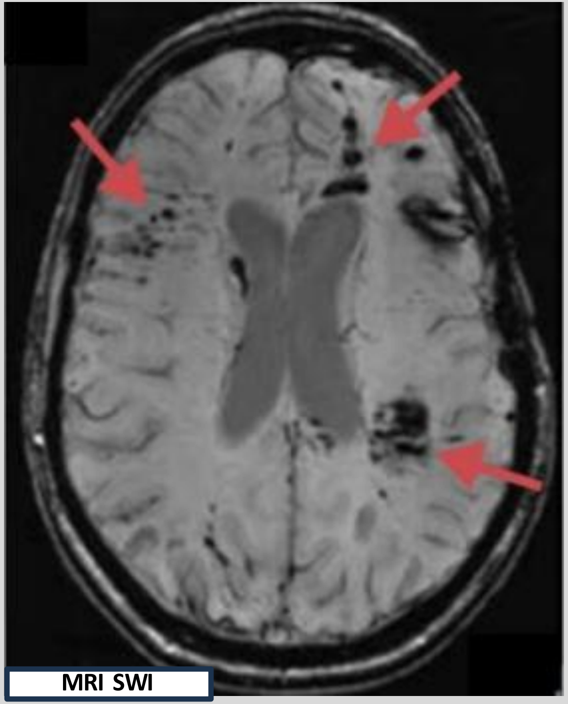
- **Most common *spontaneous*** SAH is **aneurysmal**
- Most commonly **saccular**, aka "**berry"** aneurysms of the **Circle of Willis** (in subarachnoid space)
- **Berry aneurysms** - small, thin-walled protrusions of usually arteries of circle of Willis or major branches
- **Majority** are on **anterior circle of Willis** (**anterior communicating** artery most common – see diagram)
- In ~ 2% of population; multiple in 20-30% of cases in autopsy series
- Risk factures
    - Collagen/Elastin disease (ex. **Marfan's** & **Ehlers Danlos** syndromes)
    - **Female**
    - **Family history**
    - **Smoking**
    - Autosomal dominant polycystic kidney dz
- Aneurysms are few millimeters to 2 or 3 cm in diameter, bright red, shiny surface and thin, translucent wall - risk of bleed ↑ with size: >10 mm diameter have ~50% risk of bleeding per year
- May rupture any time, but ~$1/3^{rd}$ of cases acute ↑ICP, like straining at stool or sexual orgasm
- 25-50% of patients die with first rupture, but survivors often improve and recover consciousness in minutes - repeat bleeding is common with worse prognosis each time
- Other aneurysm types include atherosclerotic (fusiform; mostly of the basilar artery), mycotic, traumatic, and dissecting but more often cause cerebral infarction than SAH



Robbins and Cotran, Pathologic Basis of Disease, 10th edition, 2021
Chapter 28 fig. 28.19

Diffuse axonal injury (DAI)

- Traumatic brain injuries **damage deep white matter regions**, cerebral peduncles, superior colliculi, and deep reticular formation in brainstem
- Widespread brain injury from **acute stretching & shearing forces** during trauma, damaging nerve fibers (**axons**)
- Microscopic findings include axonal **swelling**, and **focal hemorrhagic** lesions subsequently there's degeneration of involved fiber tracts
- Imaging studies show **petechial hemorrhages** in white matter tracts
- **High speed brain injury** may occur from marked changes in **angular acceleration** (e.g., **blast injuries, MVC**), even in **absence of physical impacts of skull**
- Associated with **immediate, deep coma**
- Cerebral **edema** and **↑ICP**
- Mortality 30-40%



- magnetom_flash_ismrm_issue_62-02161984_1800000002161984.pdf
- For detecting tiny hemorrhages (**microbleeds**) and axonal damage invisible to CT, especially in white matter, brainstem, and corpus callosum - **SWI** is highly sensitive for detecting these hemorrhagic lesions, which are key indicators of DAI
- Susceptibility-weighted magnetic resonance imaging (**MRI**) scan of the brain, highlighting **multiple cerebral microbleeds**
- Dark spots (red arrows) represent hemosiderin deposits from old, small hemorrhages